

Biology 201L: Molecular Biology

MWF 10:05-11:20 in BioSci 111

Canvas.duke.edu

Welcome! We welcome you to this course and to Duke Biology. As educators, we think about you holistically as an individual with identities different from your peers and strive to make an environment that feels inclusive. To support you as a learner, we've designed a course with lots of opportunities to practice at your own pace and with peers. Our course design follows research-based methods for how learning works. If you would like to give feedback about your learning at any time, you can fill out this [form](#) (anonymous option).

BIOL 201L Teaching Team:

Professors

Dr. Kelly Hogan (she/her)

Email: Kelly.Hogan@duke.edu

Office Location: 136 Biosciences
announced in our weekly emails.

Student office hours: by appointment and sometimes

Let's get to know each other in office hours or FLUNCH. I LOVE teaching introductory courses and love getting to know students personally. If you want to learn more about me as a person, click [here](#).

Dr. Arnaldo Carreira-Rosario (he/him)

Email: Arnaldo.Carreira@duke.edu

Office FFSC 4220A

Student office hours: by appointment

FLUNCH is always welcome!

Course/Lab Coordinator

Mr. Chris Shreve (he/him)

Email: Christopher.shreve@duke.edu

Office Location: 0054 BioSci

Student office hours: by appt. (between 10am and 5pm)

Lab Instructors	Graduate and Undergraduate Teaching Assistants	
Shannon Dougherty	Leo Camino	Mao Kobayashi
Lara Linden-High	Shreya Sarkar	Barbara Marinescu
Jola Miadlikowska	Mason Schmidt	Caroline McSwain
Becky Cary	Shannon Skarha	Sara Torres
	Sharan Surya Sekar Manjula	Larkin Woodard
Undergraduate Learning Assistants (ULAs)		
Lecture: Kate Choi	Kemmora Simmons	Avry Kosteletzky
Suraj Chatoth	Pierson Lee	Samantha Arteaga
Lab: Penelope Gaspard	Rain Xie	Kathryn Homa
Emily Pabalate	Evan Gillespie	Matthew Zhang
		Issac Jung
		Selena Xiao
		Cecilia Emerenciano Gurgel



Course lecture schedule: The [detailed lecture schedule](#) (with daily class objectives and assignments) is posted as a google doc. The schedule is subject to small changes, but exam dates will only change if there are university closures (extreme weather, etc.).



Required Resources: Modified Mastering Biology with Pearson eText for Campbell Biology in Focus; By Lisa Urry, Michael Cain, 4th edition--ISBN# 978-0-13-822521-6. The e-textbook is required and should include: 1) Mastering Biology, 2) ebook, and 3) Learning Catalytics. Purchase a unique access code via the Duke Bookstore or directly from Pearson. Follow [these instructions](#) to access the course materials.

Internal Document Links to:

[Resources for Success](#)
[Grading Overview](#)
[Lab Details, Policy, and Calendar](#)

[Lecture Course Description and Structure](#)
[Graded Components and Policies Details](#)
[Additional Campus Resources and Policies](#)

Resources for Success:

1. **Q & A: (In class and out of class).** Ask questions during class! Use the Ed Discussions feature on Canvas to ask questions and see what your classmates are asking about.
2. **Office/Student Hours with Dr. Hogan, Dr. Carreira-Rosario, and Mr. Shreve:** Please come discuss class material, personal matters, or chat about careers and more. Times and days? See Canvas homepage for details.
3. **Practice Sessions with TAs and ULAs:** Come work through the additional practice problems with peers and the Teaching Team to test your understanding and strengthen connections. See Canvas for schedule.
4. **SAGE Study Groups:** [The STEM Advancement Through Group Education \(SAGE\) program](#) is an optional resource offered in conjunction with the Academic Resource Center. These groups meet weekly with a trained undergraduate Peer Facilitator. SAGE helps students learn effective STEM study strategies, provides extra support for the challenging material in the course, and fosters a collaborative community of learners. Sign-ups will begin after the drop/add period.
5. **Peer Tutoring:** [Peer tutoring](#) is also available for this course and offered in conjunction with the Academic Resource Center.

Lecture Course Description and Structure:

Some competencies associated with our course:

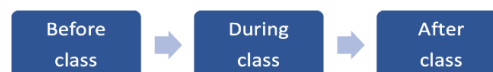
- Use and become confident with biological research thinking (design experiments with controls, predict outcomes, interpret figures, make conclusions, etc.)
- Communicate and collaborate with your peers to explain concepts and solve problems in molecular biology
- Explain and construct models associated molecular concepts and experimental data
- Reflect on your own learning process and make adjustments as needed (metacognition)

For specific learning objectives for class sessions, see the [detailed lecture schedule](#).

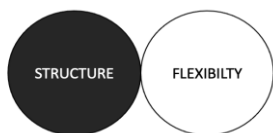
*This course fulfills the Natural World (NW) or Natural Sciences (NS) area of knowledge and the Science, Technology, and Society (STS) mode of inquiry of the Trinity College General Education requirements. **Prerequisites:** none*

Course Structure: We've designed the course with practice and feedback before, during, and after each lesson--which are important components of learning.

We have evidence that this structure ensures all learners more equitable access to being successful. Want to see these data that show that structured learning promotes equitable learning for students? Ask – we'd be happy to share!

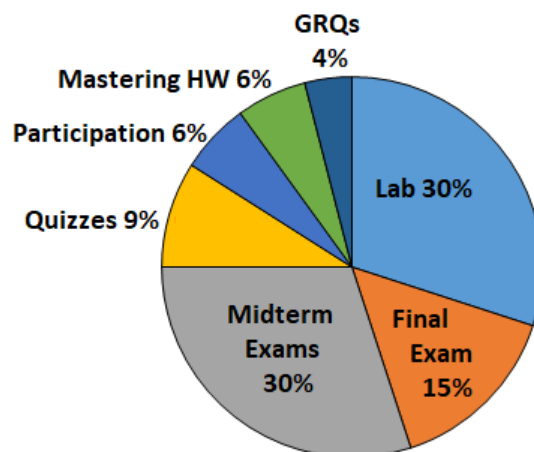


Flexibility and Growth Mindset: Sometimes life gets in the way of learning. Sometimes you make a mistake, but you can learn from it. We have built some flexibility and mistake-making into the course. You do not have to be perfect! Let go of perfectionism and try to enjoy learning for the sake of learning. Be kind to yourself and recognize when you need some self-compassion. If the built-in flexibility available to all students is not working due to something major going on in your life, please speak with Dr. Hogan or Mr. Shreve by email or in person.



Grading Overview:

- 1. Pre-class Guided Reading Questions (GRQs)** (4%): due before class, graded for completion, but not for correctness; flexibility: drop 3.
- 2. Pre-class Mastering Biology** (6%): due before class and can be re-taken a second time to improve score/learning (if still before class); flexibility: drop 3 lowest grades.
- 3. Participation** (6%) in class via group work and polls; graded 50% on correctness, 50% on participation. Flexibility: drop lowest 4 scores.
- 4. Quizzes** (9%): in person; timed; a data sheet is allowed; Flexibility: drop 1 lowest score.
- 5. 3 Midterm Exams** (30%): in person, timed, no data sheet; flexibility: drop 1 lowest score.
- 6. Cumulative Final Exam** (15%)
- 7. Lab** (30%): See "Lab Policies" below for details



Final grades

>97=A+, >93=A, >90=A- >87=B+, >83=B, >80=B- >77=C+, >73=C, >70=C-
 >67=D+, >63=D, >60=D- less than 60 = F

You may also be wondering...is there a pre-determined number of students that receive an A? Absolutely not. In theory, if the whole class earns A's, then the whole class is given A's. Earning an A is something you can achieve. It will require routine practice (built into the course structure), reflection, honesty about how your learning is progressing, and active study skills.

Graded components and policies details:

1. Pre-class Guided Reading Questions (GRQs): These are designed to help convey expectations about what to take away from **reading assignments**. They will give you practice through a question/answer format. These are not graded for correctness, and you can feel free to make them your own personalized study guides via drawings, colors, and more. Students usually find these quite helpful and, in the past, have asked for these to be required via upload

(before class) to help stay on track (**structure**). We will look to make sure they are complete, but they are not graded on content. Collaboration is allowed. You can drop 3 (**flexibility**).

How can AI help you in this course? AI can be a great study partner, especially to help quiz you on some basic concepts. For example, you can put your completed GRQs into it and say “quiz me on this material”. (A study partner is also a great way to do this). AI may also be able to help answer specific curiosity questions that arise for you—but be careful to verify the information, as it has been known to produce false answers sometimes. While AI might help you with GRQs, you’ll need to verify the information with the textbook (so maybe just use the textbook?) Additionally, GRQs are not meant to be busy work—they are meant as your first time to practice explaining and drawing a concept for yourself. If ChatGPT does this, then you won’t be as successful as *you* can be, and it will ultimately take longer for you to learn the course material on your own.

2. Pre-class reading assessment homework via Mastering Biology: These are designed to give you another chance to practice concepts and **test yourself** on your comprehension from the reading (**structure**). These are graded for correctness, but you get multiple attempts per question and you can re-do the assignment once if it is before the deadline (before class). After the due date, these can be repeated for practice. These are designed for you to assess your own learning, but collaboration is allowed. You can drop 3 (**flexibility**).

3. In class participation via group work uploads and polls: We will be doing a variety of in-class collaborative activities, often linked to our polling technology (Learning Catalytics) or Canvas site to help you learn and stay engaged (**structure**). Polling technology is used to assess learning in real time with students *inside* the classroom. As such, if you are using polling technology outside the classroom, you will not receive credit for the day. You can drop 4 class days with no explanation needed (**flexibility**). Missed a class? Class sessions will be recorded and posted, but you cannot make up the polls from a missed class unless it is part of an excused absence. Participation via polling will be graded 50% on correctness and 50% on engagement/effort, so you do not have to be perfect! We only ask that you try your best. If your **semester average** for Participation is 75% or better at the end of the semester, we will round it up to 100%! Your lowest 4 scores will be dropped (**flexibility**).

Staying engaged in a class looks different for different people. If you need to stand up during lecture, leave the room, fidget with a toy, etc. you should do so as needed. Please only use your laptop for class-related activities only. If we see you off-task (watching sports, shoe shopping, etc.) we may ask you to put your laptop away so as not to distract others in their learning.

We will work in **regular groups** during lecture, and this is so that we can learn from each other. If that will be difficult for you, please talk to Dr. Hogan or Mr. Shreve. We’ll ask that you consider a few general thoughts as we engage in group work:

- Assume best intentions of your peers (ex: if someone is not prepared for class assume there is a legitimate reason and do your best to include them)
- Take turns talking (rotate who explains, who listens, who provides clarification)
- Be clear with communication and expectations up front to avoid problems later (ex: discuss how you best work and what hasn’t worked well for you in the past)
- Get to know each other and have fun! Find out what you have in common.

Need to miss class because of illness?

If you are really sick (fever, vomiting, etc.), PLEASE don’t come to class! Take care of yourself first, and when you can, submit the [Incapacitation Form](#) to **only Mr. Shreve**; no need to copy all Bio201L professors. Once you have submitted an incapacitation form, please remember that you have 48 hours to communicate directly with Mr. Shreve about missed work. If you have

ongoing health issues, be sure to contact your academic dean, so we can work with them to keep you on track.

4. Quizzes: These will be taken in class, open-note (a “data sheet” page with your own notes on front and back). Quizzes will be taken individually on paper and will take ~20 minutes during the start of the designated class period. They are mostly multiple choice and designed to help you review material from class and prepare for exams with exam-like questions. Quizzes will be graded and returned via Gradescope in Canvas. Your one lowest quiz score will be dropped (**flexibility**). If you have accommodations for testing, please talk to Mr. Shreve.

5. Exams: Exams will be closed-notes and in-person, taken individually on paper under the Duke Community Standard (75 min for midterm exams; each mid-term exam will not be cumulative). The format will be a mix of multiple-choice type questions and short answer (ranging from one word to a few sentences). Mid-term exams will be graded and returned via Gradescope. We will drop your lowest mid-term exam (**flexibility**).

6. Cumulative Final Exam: The final exam will be cumulative and not returned via Gradescope.

Exam makeups are only given for excused absences and make up exams must generally be taken within a week through the Testing Center. Excused absences require the appropriate University form (Incapacitation Form, NOVAP, Religious Observance) or a Dean’s excuse.

If you have accommodations through the [Student Disability Access Office \(SDAO\)](#), we encourage you to use these for your success, and let us know more about your individual needs via a short in-person chat or email. If you think you may need accommodations, please contact the SDAO. We will be using the [Testing Center](#) in Trent Hall for **midterm exam** testing accommodations (but NOT for quizzes or the Final Exam). **Please be sure to schedule all your exam appointments for the semester by the end of drop/add.** (The Testing Center requests this be done at least a week before each exam).

Lab Details, Policies, and Calendar

The goal of this lab is to introduce to you the molecular basis for biology, specifically the Central Dogma of molecular biology, and to prepare you for further study in the biological sciences. The lab is organized to give each student the best opportunity to master the material in a variety of formats, including hands on activities, discussion questions, and homework assignments. By incorporating several different kinds of assessment, you will be exposed to the core concepts of molecular biology in different ways, all with the goal of enriching your understanding of the exciting field of molecular biology.

Unlike many science labs, the labs in this course do not have written lab reports! You will NOT need a carbon-paper notebook, though you will want a notebook for taking notes during lab. You also do not need to wear safety goggles or closed-toe shoes. All we expect you to bring to lab are the day’s protocol, a writing tool, and a positive attitude.

Each week, there will be a module posted on the Canvas site that contains the following:

- **Background reading** for the lab – This information gives you context for the lab exercises and provides key definitions and explanations to help you get the most out of each lab. You should read this carefully, but you do not need to bring a copy to lab.
- **Protocol** – This document details the procedure you will perform in the lab. You should

read it carefully before lab AND print a copy and bring it with you to lab.

- **Pre-lab assignment** – These assignments are your chance to check your understanding of the Background and Protocol before lab. You should not need to do any extra research to answer these questions. Pre-lab assignments are due at the start of lab period. There will also be a short Pre-lab “check-in” quiz at the start of each lab. We will automatically drop everyone’s one lowest pre-lab score.
- **Post-lab assignment** – These assignments will incorporate concepts from the lab activity, the in-lab practice problems, and the related Background. Post-labs should be completed and turned in at the start of the next lab period (e.g., the post-lab for lab 3 is due at the start of lab 4). Post-labs finished during the next lab or turned in after the lab period will be assigned a 10% late penalty. Post-labs will not be accepted past 24 hours after they are due. We will automatically drop everyone’s one lowest post-lab score.

You are free to talk with one another and gather information about the pre- and post-lab assignments from various sources (including TAs/Lab Instructors); however, you need to do the write-up of the pre- and post-lab assignments for yourself. We want to know that when we evaluate your answers on an assignment, that the assignment represents the words and expression of YOUR understanding. You should always be able to explain the logic and thinking behind any answer you submit. It might be tempting to use AI to answer the homework problems, but if you let AI do the thinking for you, you won’t learn the material as well, and lab material shows up on quizzes and exams! It can also be tempting to get answers from inappropriate sources – previous semester’s assignments or copying from a classmate this semester – but remember that homework is designed to help *you* to learn the material. If we suspect these inappropriate sources (AI or copying) were used for homework, we reserve the right to give assignments that violate our course policies a zero that will not be dropped.

Lab Participation and Professionalism:

Your participation grade is a score awarded to you each week by your Lab Instructor/TA based on the extent to which you participated in lab activities in a constructive manner. Missing or coming late to lab, texting during lab, failing to follow directions, or otherwise not giving your classmates and the lab work your full attention during lab period will adversely affect your participation score. Helping your lab-mates, answering and asking thoughtful questions, and carefully following directions will enhance your participation score. If you have concerns about your score or feel that your TA is being unfair, please talk with Mr. Shreve.

Student Conference:

At the end of the semester, you will have a conversation with your Instructor/TA summarizing the results of your cloning project and what you learned about the DNA sequence you isolated. We will provide more detailed guidelines toward the end of the semester.

Attendance

Weekly attendance at your scheduled lab is expected. **Please arrive at the lab room a few minutes early so that we can start (and end) on time!** The lab sections will consist of some cooperative and interactive activities among the students in each section; if you are late, it can delay or disrupt these activities and make it difficult for everyone to finish on schedule. Lateness will also count against your participation score for the day!

Anticipated Absences

Notification of all anticipated lab absences, such as religious holidays or varsity sports*, should be submitted to Mr. Shreve (Christopher.shreve@duke.edu) by the end of the drop/add period.

For religious holidays, see [here](#); For varsity sports, see [here](#).

*Note that only absences for varsity sports are excused. Club sports are not eligible.

Unanticipated Absences

For unanticipated **lab absences**, send the [Incapacitation Form](#) to Mr. Shreve before your absence or as soon as possible afterward. In addition to the official University form, you **MUST** also complete the [Lab Absence Notification form](#) found on the course Canvas page. This form allows us to find a seat in another lab that week for you to make up the missed lab activity. We will do our best to work with you to keep you caught up with the lab material. Our amazing prep staff will also be sure to have a replacement sample ready for you if you can't make up lab during the week. Labs do not meet on Friday, and it is not possible to make up a lab exercise after the week it is scheduled, so early notice is essential!

If you miss a lab due to illness or other absence, you are still responsible for the material covered and for completing the homework assignments. If you are rescheduled for a lab period later in the week, the relevant pre-lab assignment will be due at the start of the make-up lab. Your post-lab assignment should be submitted to your TA (in person or by email) by 5pm the day **after** your regularly scheduled lab, regardless of when you make up lab.

If you're having trouble keeping up with the lab assignments, please come talk with us! We want you to succeed and will work with you and your academic dean to provide the best support we can. We've helped students in all kinds of circumstances, so don't be afraid to ask for help.

Lab Calendar for Spring 2026

Week of Aug 24: Lab 0 – Getting to know you!

Week of Aug 31: Lab 1 – Introduction to yeast as a model system

No labs the week of Sept 7 (Labor Day Monday Holiday)

Week of Sept 14: Lab 2 – DNA extraction from Yeast Cells

Week of Sept 21: Lab 3 – DNA digestion and gel electrophoresis

Week of Sept 28: Lab 4 – Ligation and Transformation

Week of Oct 5: Lab 5 – Plasmid extraction and PCR setup

No labs the week of Oct 12 – Fall Break!

Week of Oct 19: Lab 6 – Restriction Digestion and Gel Electrophoresis

Week of Oct 26: Lab 7 – Restriction Mapping and DNA Sequencing

Week of Nov 2: Lab 8 – Introduction to Bioinformatics

Week of Nov 9: Lab 9 – Sequence Analysis of Yeast Genomic Library

Week of Nov 16: End of Semester Conferences

No labs the week of Nov 23 (Thanksgiving Break)

Week of Nov 30: End of Semester Conferences

Acknowledgements: We want to thank Dr. Alison Hill for her development of the lab curriculum.

Additional Campus Resources and Policies

Technology and Digital Resource Assistance

Highly aided students who have limited access to computers may request loaner laptops through the [LIFE Loaner Laptop Program](#). Please note that supplies are limited.

If you are having difficulty with textbook and supply costs associated with this course, contact the [financial aid office](#) (whether or not you are on aid) to receive aid or a loan or contact [Duke LIFE Course Materials Assistance](#) to see if you are eligible.

Students with disabilities

If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the Student Disability Access Office (SDAO) and provide them with documentation of your disability. SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations are not retroactive and disability accommodations cannot be provided until a Faculty Accommodation Letter has been sent to Mr. Shreve. Contact SDAO for more information: sdao@duke.edu or www.access.duke.edu.

Not a native English speaker/reader? If you are struggling, come talk to Dr. Hogan so we can find ways to ensure your success.

Lectures will be recorded in this course. Recordings of lectures are permitted for private study only. You cannot distribute lecture recordings to anyone else without our authorization. Unauthorized distribution of course recordings or other materials is a cause for disciplinary action.

Health and Wellness: If you are concerned about your physical or mental health, [DukeReach](#) can connect you with departments across campus to get you help, and you can contact [Counseling & Psychological Services \(CAPS\)](#) directly for counseling services. You can also reach 24/7 mental health support through [TimelyCare](#).

Academic Integrity

As a student, you should abide by Duke's [Community Standard](#), which states: Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect and accountability. Citizens of this community commit to reflect upon and uphold these principles in all academic and nonacademic endeavors, and to protect and promote a culture of integrity.

To uphold the Duke Community Standard:

- I will not lie, cheat, or steal in my academic endeavors.
- I will conduct myself honorably in all my endeavors; and
- I will act if the Standard is compromised.

It is the responsibility of the student to understand and follow Duke policies regarding academic integrity, including doing one's own work, following proper citations of sources, and adhering to guidance around group work. Please speak with me to get any clarification about this course.